Suozhi Huang

Ph.D. Student in Electrical and Computer Engineering, Princeton University

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RESEARCH PROFILE

Ph.D. student at Princeton University working on AI for Science, world models, post-training, and agentic systems, with an emphasis on reliable decision-making in dynamic environments. Seeking 2027 internship opportunities.

EDUCATION

Princeton University — Princeton, NJ
Ph.D. Student, Electrical and Computer Engineering; Advisor: Prof. Mengdi Wang — *Sep 2025 – Present*

Tsinghua University — Beijing, China
B.E., Computer Science and Technology (Artificial Intelligence), Yao Class — *Sep 2020 – Jun 2025*

New York University — New York, NY
Exchange Student, Tandon School of Engineering — *May 2023 – Sep 2023*

RESEARCH APPOINTMENTS

Princeton University — Princeton, NJ
Graduate Researcher; Advisor: Prof. Mengdi Wang — *Sep 2025 – Present*

Shanghai AI Laboratory — Shanghai, China
Research Intern, Formal Mathematical Reasoning; Advisors: Dr. Zheng Yuan and Prof. Dahua Lin — *Jun 2024 – Oct 2024*

Anima AI+Science Lab, Caltech — Pasadena, CA
Researcher, Neural Theorem Proving; Advisor: Prof. Anima Anandkumar — *Aug 2024 – Jan 2025*

New York University — New York, NY
Visiting Researcher, Collaborative Perception; Advisors: Prof. Chen Feng and Dr. Yiming Li — *Feb 2023 – Sep 2023*

RESEARCH PROJECTS AND PUBLICATIONS

LabWorld: Learning Interactive 4D Worlds for Scientific Experimentation
Suozhi Huang*, Kelsey Fu*, Binxu Li, Luna Lyu, Kaize Wu, Wenbo Lu, et al.
arXiv:2610.36592, 2026. *Paper* | *Project* | *Code*

- Developed digital twins and 4D scene reconstructions of scientific laboratory environments, including trajectory construction for robot simulation training data.

AQuA: Recursively Self-Improving Quantitative Trading Research Agents
Jiacheng Guo, **Suozhi Huang**, Yunlong Gao, Zihao Li, Jason Ge, Xu Kuang, Mengdi Wang
arXiv:2608.12841, 2026. *Paper*

- Built the data and evaluation framework, connecting financial data with assessments of agents' financial analysis and quantitative research capabilities.
- Designed and implemented the self-improving agent harness with persistent memory, enabling iterative research and experimentation informed by prior findings and evaluation feedback.

Gated Attention for Large Language Models: Non-linearity, Sparsity, and Attention-Sink-Free
Zihan Qiu, Zekun Wang, Bo Zheng, et al., **Suozhi Huang**, et al.
NeurIPS 2025, ***Best Paper Award****.* *Paper*

- Participated in discussions on the potential role of attention gating in large language models.

InternLM2.5-StepProver: Advancing Automated Theorem Proving via Critic-Guided Search
Zijian Wu*, **Suozhi Huang***, Zhejian Zhou, Huaiyuan Ying, Zheng Yuan, Wenwei Zhang, Dahua Lin, Kai Chen
ICML 2025 AI4Math Workshop, ***Best Paper Honorable Mention (Runner-Up)****.* *Paper* | *Code*

- Built a critic model for expert iteration on Lean Workbook and Lean GitHub; replaced baseline best-first tactic selection with critic-guided search.
- Constructed a new proof dataset through expert-iteration search, contributing to the system's 65.9% miniF2F-test pass rate.

LeanProgress: Guiding Search for Neural Theorem Proving via Proof Progress Prediction
Suozhi Huang, Peiyang Song, Robert Joseph George, Anima Anandkumar
First author of the original arXiv v1/v2 submission; subsequently published in TMLR, 2025. Original version | TMLR

- Generated proof-search tree data and trained and validated models to predict remaining proof steps and proof progress.
- Integrated progress-prediction functionality into LeanCopilot to provide additional guidance during interactive theorem proving.

CryptoBench: A Dynamic Benchmark for Expert-Level Evaluation of LLM Agents in Cryptocurrency
Jiacheng Guo*, **Suozhi Huang***, Zixin Yao*, Yifan Zhang*, Yifu Lu*, et al.
arXiv:2512.00417, 2025. Paper

- Designed financial analysis tasks, collected benchmark data, and constructed reference answers and scoring rubrics for on-chain analysis, retrieval, and prediction.
- Built the agent framework and conducted evaluation experiments across direct and agentic settings, assessing 10 leading LLMs on expert-level financial tasks.

Agentic Laboratories of the Future: Towards World Models for Scientific Discovery
Kelsey Fu, Luna Lyu, Shuzhen Li, **Suozhi Huang**, et al.
Preprints.org, 2026. Preprint (not peer-reviewed). Paper

- Primarily responsible for digital twins and 4D scene reconstruction of scientific laboratory environments.

Any2Poster: Any-Source Poster Generation Across Modalities and Domains
Amogh Vinaykumar, Aiden Li, **Suozhi Huang**, Shilong Liu
arXiv:2606.02915, 2026. Paper | Code

FutureX-Pro: Extending Future Prediction to High-Value Vertical Domains
Jiashuo Liu, Siyuan Chen, Zaiyuan Wang, Zhiyuan Zeng, Jiacheng Guo, Liang Hu, Lingyue Yin, **Suozhi Huang**, et al.
Princeton–ByteDance Seed collaboration; arXiv:2601.12259, 2026. Paper

- Built the FutureX-Finance dataset and benchmark, covering 150 companies across U.S. and Chinese A-share markets.
- Constructed tasks for single-day prices, multi-day price extrema, and directional momentum to evaluate LLM agents on future market outcomes.

ActFormer: Scalable Collaborative Perception via Active Queries
Suozhi Huang*, Juexiao Zhang*, Yiming Li, Chen Feng
IEEE International Conference on Robotics and Automation (ICRA), 2024. Project

- Proposed active query-based camera-only collaboration and implemented pose-guided selection in deformable attention to reduce BEV queries and feature-transfer costs.
- Adapted V2X-Sim to MMDetection3D and conducted the multi-agent experiments and result analysis to improve scalability.

A Knowledge–Data Dual-Driven Framework for Predicting Rechargeable Battery Electrolyte Properties
Yuchen Gao, Yu-Hang Yuan, **Suozhi Huang**, et al.
Angewandte Chemie International Edition, 2025. Paper

- Applied the pretrained Uni-Mol model to predict electrolyte safety properties.
- Implemented molecular subgraph interpretation using Monte Carlo tree search and Shapley values; ran baselines for a molecular-property vector database.

** Equal contribution. Full publication list available on Google Scholar.*

TECHNICAL SKILLS

Programming	Python, C++, Lean 4, Coq
Frameworks	PyTorch, OpenCV, MMDetection3D, LeanDojo
Systems and Tools	Linux, Git

HONORS

NeurIPS 2025 Best Paper Award (coauthor) 2025
Best Paper Honorable Mention (Runner-Up), ICML AI4Math Workshop 2025
Gold Award, 36th Chinese Physics Olympiad (Top 30 nationally) 2019